

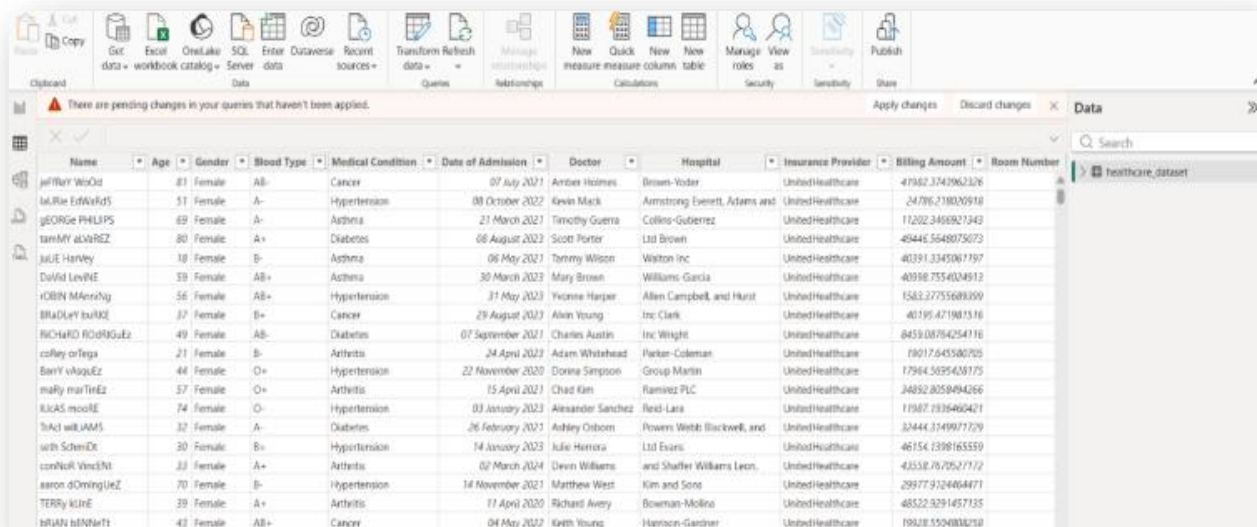
WEEK 5 – Healthcare Dataset

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INTRODUCTION

The healthcare dataset was imported into Power BI and opened in Power Query for data preparation. It includes details such as patient information, medical conditions, admission records, hospital details, insurance providers, billing amounts, medications, and test results.

Before starting the cleaning and transformation process, the dataset was reviewed to get a clear understanding of the patient, admission, and billing-related fields.



Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number
Jeffrey West	81	Female	A-	Cancer	07 July 2021	Amber Holmes	Brown-Voder	UnitedHealthcare	47982.3743962326	
Julia Edwards	51	Female	A-	Hypertension	08 October 2022	Kevin Mack	Armstrong Everett, Adams and	UnitedHealthcare	24786.278820918	
George Phillips	69	Female	A-	Asthma	21 March 2021	Timothy Guerra	Collins-Gutierrez	UnitedHealthcare	11202.346821343	
Tammy Alvarez	80	Female	A+	Diabetes	08 August 2023	Scott Porter	Litt Brown	UnitedHealthcare	49446.5648075073	
Julie Harvey	18	Female	B-	Asthma	06 May 2021	Tommy Wilson	Walton Inc	UnitedHealthcare	40391.3345067197	
David Levine	59	Female	A+	Asthma	30 March 2023	Mary Brown	Williams Garcia	UnitedHealthcare	40998.7554024912	
Robin Marsh	56	Female	A+	Hypertension	31 May 2023	Yvonne Harper	Allen Campbell, and Hurst	UnitedHealthcare	1583.27755689399	
Bradley Burke	37	Female	B+	Cancer	29 August 2023	Alan Young	Inc Clark	UnitedHealthcare	40195.471981516	
Richard Rodriguez	49	Female	A-	Diabetes	07 September 2021	Charles Austin	Inc Wright	UnitedHealthcare	8459.08764254716	
Colley Ortega	21	Female	B-	Arthritis	24 April 2023	Adam Whitehead	Parker-Coleman	UnitedHealthcare	19017.645580705	
Barry Valquez	44	Female	O+	Hypertension	22 November 2020	Dorena Simpson	Group Martin	UnitedHealthcare	17964.5095426175	
Emily Martinez	57	Female	O+	Arthritis	15 April 2021	Chad Kim	Ramirez PLC	UnitedHealthcare	24892.8058494266	
Eliza Moore	74	Female	O-	Hypertension	03 January 2023	Alexander Sanchez	Ried-Lara	UnitedHealthcare	11987.7336460421	
Isaac Williams	32	Female	A-	Diabetes	26 February 2021	Ashley Osborn	Powers Webb Blackwell, and	UnitedHealthcare	32444.3149977729	
Wendy Schmidt	30	Female	B+	Hypertension	14 January 2023	Julie Herrera	Litt Evans	UnitedHealthcare	46154.1398165559	
Conor Vincent	33	Female	A+	Arthritis	02 March 2024	Devin Williams	and Shaffer Williams Leon	UnitedHealthcare	43538.7670527172	
Saron Dominguez	70	Female	B-	Hypertension	14 November 2021	Matthew West	Kim and Song	UnitedHealthcare	29977.910464471	
Terry Kline	39	Female	A+	Arthritis	17 April 2020	Richard Avery	Bowman-Molina	UnitedHealthcare	48522.9291457135	
Brian Martin	43	Female	A+	Cancer	04 May 2022	Kath Young	Harrison-Gardner	UnitedHealthcare	19828.550488258	

Cleaning Data

The categorical columns were cleaned using the **Trim** and **Clean** transformations. This helped remove extra spaces and unwanted characters, making the text values more consistent and reliable for analysis.

The **Name** column was also standardized using **Capitalize Each Word**, so that names followed a uniform capitalization format.

Power Query Editor interface showing a table with columns: Billing Amount, Room Number, Admission Type, Discharge Date, Medication, and Test Results. The 'Test Results' column is highlighted, and the 'Capitalize Each Word' option is selected in the 'Format' menu.

Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
18816.28242	328	Urgent	02-02-2020	Paracetamol	Normal
32643.32729	265	Emergency	26-05-2019	Ibuprofen	Inconclusive
27953.09608	305	Emergency	07-10-2022	Aspirin	Normal
27905.78242	450	Elective	18-12-2020	Ibuprofen	Abnormal
14238.82781	458	Urgent	09-10-2022	Penicillin	Abnormal
48145.11095	389	Urgent	24-12-2022	Ibuprofen	Normal
19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
50119.22279	328	Elective	14-07-2020	Aspirin	Abnormal
19784.63306	249	Elective	22-06-2021	Paracetamol	Inconclusive
12576.79581	394	Urgent	23-04-2020	Paracetamol	Normal
7995.58888	288	Urgent	05-09-2022	Lipitor	Inconclusive
43282.28386	134	Emergency	28-12-2019	Penicillin	Abnormal
25207.70863	808	Urgent	19-06-2020	Lipitor	Normal
40701.58823	382	Urgent	13-10-2021	Paracetamol	Normal
12263.35783	489	Elective	12-01-2022	Aspirin	Inconclusive
24499.8479	124	Elective	14-07-2020	Paracetamol	Normal
17483.46544	449	Urgent	02-06-2020	Paracetamol	Abnormal
18845.62852	280	Elective	14-03-2021	Aspirin	Abnormal

Creating Age Group

An **Age Group** column was created using a Conditional Column in Power Query. The patients were divided into four age categories:

- 0–18
- 19–35
- 36–60
- 60+

This grouping makes it easier to study patients according to different age ranges.

Power Query Editor interface showing a table with columns: Age, Gender, Medical Condition, Date of Admission, and Doctor. The 'Age' column is highlighted, and the 'Capitalize Each Word' option is selected in the 'Format' menu.

Age	Gender	Medical Condition	Date of Admission	Doctor
80	Male	Cancer	01-01-2024	Matthew Smith
62	Male	Obesity	20-08-2019	Samantha Davies
76	Female	Obesity	22-09-2022	Triffany Mitchell
28	Female	Diabetes	18-11-2020	Kevin Wells
42	Female	Cancer	19-09-2022	Kathleen Hanna
85	Male	Asthma	20-12-2024	Taylor Newton
22	Female	Diabetes	02-11-2020	Kelly Olson

	Valid	Error	Empty
1	Bobby Jackson		
2	Leslie Terry		
3	Danny Smith		
4	Andrew Watts		
5	Adrienne Bell		
6	Emily Johnson		
7	Edward Edwards		
8	Christina Martinez		
9	Jasmine Aguilar		
10	Christopher Berg		
11	Michelle Daniels		
12	Aaron Martinez		
13	Connor Hansen		

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name:

	Column Name	Operator	Value		Output
If	Age	is less than or equal to	18	Then	0-18
Else If	Age	is less than or equal to	35	Then	19-35
Else If	Age	is less than or equal to	60	Then	36-60

Creating Dim_Patient

A separate **Dim_Patient** table was created by keeping the main patient-related fields. It includes **Name, Age, Gender, Blood Type, and Age Group**.

A **Patient ID** was added to uniquely identify each patient record and establish a connection between patient information and the Billing table.

Creating Dim_Admission

The **Dim_Admission** table was created to organize the admission-related information separately. It contains **Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, and Test Results**.

An **Admission ID** was assigned to identify individual admission records and link them with the Billing table.

Columns	Function	From Text	From Number	From Date & Time
General				

Table.SelectColumns(*Reordered Columns*,{"Name", "Age", "Gender", "Blood Type", "Age Group"})										Query Settings
Dim_Patient	Name	Age	Gender	Blood Type	Age Group					PROPERTIES
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%					Name
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%					Dim_Patient
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%					All Properties
1	Bobby Jackson	30	Male	B-	19-35					APPLIED STEPS
2	Leslie Terry	62	Male	A+		60				Source
3	Danny Smith	76	Female	A-		60				Promoted Headers
4	Andrew Watts	28	Female	O+	19-35					Changed Type
5	Adrienne Bell	49	Female	AB+	36-60					Trimmed Text
6	Emily Johnson	36	Male	A+	36-60					Cleaned Text
7	Edward Edwards	22	Female	AB-	19-35					Trimmed Text1
8	Christina Martinez	20	Female	A+	19-35					Cleaned Text1
9	Jasmine Aguilar	32	Male	AB+		60				Trimmed Text2
10	Christopher Berg	58	Female	AB-	36-60					Cleaned Text2
11	Michelle Daniels	72	Male	O+		60				Trimmed Text3
12	Aaron Martinez	38	Female	A-	36-60					Cleaned Text3
13	Connor Hansen	75	Female	A+		60				Trimmed Text4
14	Robert Bauer	68	Female	AB+		60				Cleaned Text4
15	Brooke Brady	44	Female	AB+	36-60					Trimmed Text5
16	Mrs. Natalie Gamble	46	Female	AB-	36-60					Cleaned Text5
17	Wiley Perkins	69	Female	A+		60				Capitalized Each Word
18	Mrs. Jamie Campbell	38	Male	AB-	36-60					Trimmed Text6
19	Luke Burgess	34	Female	A-	19-35					Cleaned Text6
20	Daniel Schmidt	63	Male	B+		60				Added Conditional Columns
21	Timothy Burns	47	Female	A-		60				

Table.RemoveColumns(*Reordered Columns*,{"Name", "Age", "Gender", "Blood Type", "Age Group", "Date of Admission", "Insurance Provider"})										Query Settings
Dim_Billing	Medical Condition	Doctor	Hospital	Room Number	Admission Type	Medication	Test Results			PROPERTIES
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%			Name
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%			Dim_Admission
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%			All Properties
1	Cancer	Matthew Smith	Sons and Miller	328	Urgent	Paracetamol	Normal			APPLIED STEPS
2	Obesity	Samantha Davies	Kim Inc	265	Emergency	Ibuprofen	Inconclusive			Source
3	Obesity	Tiffany Mitchell	Cook Plc	203	Emergency	Aspirin	Normal			Promoted Headers
4	Diabetes	Kevin Wells	Hernandez Rogers and Vang	450	Elective	Ibuprofen	Abnormal			Changed Type
5	Cancer	Kathleen Hanna	White-White	458	Urgent	Penicillin	Abnormal			Trimmed Text
6	Asthma	Taylor Newton	Nunes-Murphy	309	Urgent	Ibuprofen	Normal			Cleaned Text
7	Diabetes	Kelly Olson	Group Middleton	389	Emergency	Paracetamol	Inconclusive			Trimmed Text1
8	Cancer	Isabelle Thomas	Howell Robinson and Valdez	377	Emergency	Paracetamol	Inconclusive			Cleaned Text1
9	Asthma	Daniel Ferguson	Sons Rich and	328	Elective	Aspirin	Abnormal			Trimmed Text2
10	Cancer	Heather Gay	Padilla Walker	249	Elective	Paracetamol	Inconclusive			Cleaned Text2
11	Cancer	John Denson	Schaeffer Porter	394	Urgent	Paracetamol	Normal			Trimmed Text3
12	Hypertension	Douglas Mayo	Lyons Blair	288	Urgent	Lipitor	Inconclusive			Cleaned Text3
13	Diabetes	Kenneth Fletcher	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal			Trimmed Text4
14	Asthma	Theresa Freeman	Rivera-Gutierrez	909	Urgent	Lipitor	Normal			Cleaned Text4
15	Cancer	Roberta Stewart	Morris-Jewell	182	Urgent	Paracetamol	Normal			Trimmed Text5

Merging with Dim_Patient

The Billing table was merged with the **Dim_Patient** table using the matching patient information. A **Left Outer Join** was used so that all records from the Billing table were retained.

After the merge, the **Patient ID** was expanded into the Billing table. This created a direct link between the billing records and the patient information.

Final Billing Table

The final Billing table was arranged with the main fields needed for analysis, including **Date of Admission**, **Insurance Provider**, **Billing Amount**, **Discharge Date**, **Patient ID**, and **Admission ID**.

The table is now structured and ready to be used in the Power BI data model and for dashboard analysis.

Final Data Model

The completed data model consists of three main tables: **Dim_Patient**, **Dim_Admission**, and **Billing**.

The dimension tables are connected to the Billing table through **Patient ID** and **Admission ID**. This setup keeps patient and admission details separate from the main billing records while maintaining the required relationships for analysis.

This structure provides a cleaner and more organized foundation for building the healthcare dashboard.

Healthcare Analytics Dashboard

The healthcare dashboard was built using the cleaned and transformed dataset. It gives an overall view of patient demographics, admission patterns, medical conditions, and billing information.

The dashboard includes:

- Total Patients
- Total Billing Amount
- Patients by Age Group
- Patients by Gender
- Patients by Admission Type
- Patients by Medical Condition
- Billing Amount by Medical Condition
- Billing Amount by Insurance Provider
- Patients by Test Results

Table: RenameColumns(*Added Index,{{"Index", "Admission ID"}})									
Index	Hospital	Room Number	Admission Type	Medication	Test Results	Admission ID	Query Settings		
100%	Value	100%	Value	100%	Value	100%	Name		
0%	Error	0%	Error	0%	Error	0%	Dim_Admission		
0%	Empty	0%	Empty	0%	Empty	0%	All Properties		
1	Sons and Miller	338	Urgent	Paracetamol	Normal	1	APPLIED STEPS		
2	Kim Ho	265	Emergency	Ibuprofen	Inconclusive	2	Changed Type		
3	Cook PLC	209	Emergency	Aspirin	Normal	3	Trimmed Text		
4	Hernandez Rogers and Vang	456	Elective	Ibuprofen	Abnormal	4	Cleaned Text		
5	White-White	456	Urgent	Paracetamol	Abnormal	5	Trimmed Text1		
6	Nunes-Humphrey	389	Urgent	Ibuprofen	Normal	6	Cleaned Text1		
7	Group Middleton	589	Emergency	Paracetamol	Inconclusive	7	Trimmed Text2		
8	Powell-Robinson and Valdes	277	Emergency	Paracetamol	Inconclusive	8	Cleaned Text2		
9	Sons-Rich and	338	Elective	Aspirin	Abnormal	9	Trimmed Text3		
10	Hudila-Walker	249	Elective	Paracetamol	Inconclusive	10	Cleaned Text3		
11	Schwartz-Porter	334	Urgent	Paracetamol	Normal	11	Trimmed Text4		
12	Lyons-Blair	268	Urgent	Lipitor	Inconclusive	12	Cleaned Text4		
13	Powers-Miller, and Flores	234	Emergency	Paracetamol	Abnormal	13	Trimmed Text5		
14	Rivers-Gutierrez	308	Urgent	Lipitor	Normal	14	Cleaned Text5		
15	Morris-Arellano	282	Urgent	Paracetamol	Normal	15	Capitalized Each Word		
16	Cline-Williams	483	Elective	Aspirin	Inconclusive	16	Trimmed Text6		
17	Carantes-Wells	234	Elective	Paracetamol	Normal	17	Cleaned Text6		
18	Torres, and Harrison Jones	443	Urgent	Paracetamol	Abnormal	18	Admitted Conditional Columns		

CONCLUSION

The healthcare dataset was successfully cleaned and transformed using **Power Query in Power BI**. The data was standardized, an **Age Group** category was introduced, and the information was organized into **Dim_Patient**, **Dim_Admission**, and **Billing** tables.

The use of **Patient ID** and **Admission ID** helped establish clear relationships between the tables and create a structured data model.

The completed healthcare dashboard brings together useful information about patient demographics, admission types, medical conditions, test results, and billing. Overall, the structured model makes the dataset easier to explore and provides a solid base for visualization and reporting in Power BI.

